

Student ID 1 & 2: (Do this in pairs)

Word Break rules for this worksheet:

- You are given a string s and a dictionary `dict`.
- Decide whether s can be segmented into a sequence of dictionary words.
- The segmentation must cover the *entire* string.
- Dictionary words may be reused.

1. Warm-up: yes or no?

For each pair (s, dict) , write **Yes** if segmentation is possible and **No** otherwise.

String s	Dictionary <code>dict</code>	Yes/No
tacocat	{taco, cat}	
robotdance	{robot, dance}	
catsanddog	{cats, dog, sand, and, cat}	
applepenapple	{apple, pen}	

2. Fill a Word Break table

Use $s = \text{applepenapple}$ and `dict = {apple, pen}`. Fill the WB row with T/F values for $i = 0 \dots 13$.

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13
WB														

What value do you get for `WB[13]`? Explain in one segmentation sentence.

3. Choose a DP state

Let $n = |s|$. Define a 1D DP array `WB[0..n]`.

(a) Complete the state definition:

`WB[i]` = _____

(b) What `WB[i]` correspond to the empty string and what is its value?

(c) Why does checking all split points $j < i$ make sense for this state?

